

Pre-Registered Predictions for the Entropy Flow Cosmology (EFC) Model

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Euclid prediction: [DOI 10.6084/m9.figshare.31990053](https://doi.org/10.6084/m9.figshare.31990053)
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BAO regime: [DOI 10.6084/m9.figshare.31230703](https://doi.org/10.6084/m9.figshare.31230703)
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Freeze hash (SHA-256):

dbc737979d8a773f56c620759ddded38139157fd9a2dbb321a8ffdb6e68135d2

This hash seals the Boltzmann pipeline outputs. It is independently archived at Figshare (DOI above) and is not subject to modification.

Abstract. This document pre-registers the complete set of observational predictions derived from Energy-Flow Cosmology (EFC) prior to the release of Euclid DR1 (scheduled October 21, 2026), Rubin DP2 (July–September 2026), and Simons Observatory $\times$ Euclid E_G cross-correlations (~ 2027). All numerical values are frozen, anchored in independently archived sources with persistent DOIs, and are not subject to post-hoc adjustment.

1 Model Definition

The EFC model is defined by the relativistic action derived in DOI 31876324, coupling an entropy-flow scalar field ϕ non-minimally to gravity:

$$S_{\text{EFC}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} M_{\text{Pl}}^2 F(\phi) R - \frac{1}{2} K(\rho) g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) - \lambda (\Box \phi - \Gamma(\rho)) \right] \quad (1)$$

where $F(\phi) = 1 + \alpha_\phi \phi$ with $|\alpha_\phi| \ll 1$ is the non-minimal coupling, $K(\rho)$ is the density-dependent kinetic stiffness, and λ enforces the flow constraint via a Lagrange multiplier.

Tensor sector

- Gravitational-wave speed: $c_T = c$ (exact), satisfying GW170817.
- Tensor-speed excess: $\alpha_T = 0$.

Partial EFT mapping

The following Bellini–Sawicki α -functions are defined via the entropy sigmoid $S(a) = [1 + \exp(-(\ln a - \ln a_t)/\Delta)]^{-1}$:

$$\alpha_M(a) = M_0 S(a) \quad (2)$$

$$\alpha_B(a) = B_0 \frac{dS}{d \ln a} \quad (3)$$

$$\alpha_T = 0 \quad (4)$$

These are the explicit EFT functions used in the Boltzmann integration (Section 3.2). The remaining function $\alpha_K = 1.0$ is set to a fiducial value; a full derivation from the EFC action is in progress.

2 Observables

The following effective gravitational observables are computed directly from the EFC perturbation equations (DOI 31876324, conformal Newtonian gauge):

- $\mu(k, z)$: effective Newton coupling modification.
- $\Sigma(k, z)$: lensing potential modification.
- $\eta(k, z) = \Phi/\Psi$: gravitational slip.

3 Pre-Registered Predictions (Frozen)

3.1 Survival valley (phenomenological regime)

The perturbation analysis of the EFC action yields a stable observational regime (the “survival valley”) characterised by:

$$\mu(z) < 1 \quad (\text{representative value } \approx 0.94) \quad (5)$$

$$\Sigma(z) > 1 \quad (\text{representative value } \approx 1.05) \quad (6)$$

$$\eta(z) \approx 1.10 \quad (\text{positive gravitational slip, } z \sim 0.5\text{--}1) \quad (7)$$

These values represent the typical prediction within the stable parameter region identified in DOI 31876324, not a single-run output. The combination $\mu < 1$, $\Sigma > 1$, $\eta > 1$ is EFC’s primary differentiator from $f(R)$ models (which predict $\mu > 1$) and from Λ CDM ($\eta = 1$).

3.2 Boltzmann pipeline outputs (hi_class 3.0 + efc_logistic)

Using `hi_class` 3.0 with the custom `efc_logistic` Horndeski patch (DOI 31990053), the following shifts relative to Λ CDM at the benchmark point are sealed. All shifts are evaluated relative to a Λ CDM baseline with identical background cosmology (Planck 2018 best-fit). The reference scales $k = 0.05 h/\text{Mpc}$ and $\ell = 66$ correspond to quasi-linear scales probed by Euclid DR1.

ID	Observable	Shift	Falsifiable by
P1	σ_8	+1.21%	Euclid DR1 growth (RSD/clustering)
P2	$P(k=0.05 h/\text{Mpc}, z=0.5)$	+2.09%	Euclid DR1 galaxy clustering
P3	$C_\ell^{\phi\phi} (\ell = 66)$	−6.01%	Weak-lensing / CMB lensing
P4	$E_G (\ell = 66)$	−3.98%	Galaxy-lensing + RSD joint analysis
P5	$C_\ell^{TT} (\text{ISW}, \ell = 30)$	+1.88%	Planck low- ℓ TT / ISW-LSS

Benchmark parameters (fixed):

B_0	0.02	a_t	0.30	α_K	1.0
M_0	0.06	Δ	0.30	α_T	0.0

Stability conditions: $M_0 \geq 3 B_0$; $c_s^2 > 0$; $Q_S > 0$ (satisfied in the benchmark region).

3.3 Growth sector ($f\sigma_8$)

From the leave-one-out analysis (DOI 31332730): published result $\alpha = -1.00 \pm 0.46$ (2.20σ).

Two frozen solutions (Symbiose MCMC):

- $\alpha = -0.689$ (SHA 7a850cfa)
- $\alpha = -0.702$ (SHA dbccda15)

3.4 BAO regime consistency

From the regime-transition fit to DESI DR2 BAO (DOI 31230703):

- $\Delta\text{AIC} = -18.01$ vs. ΛCDM
- $\chi^2_{\text{EFC}} = 9.33$ vs. $\chi^2_{\Lambda\text{CDM}} = 31.34$

4 Kill Criteria (Falsification Conditions)

EFC is considered falsified if *any* of the following are observed at $\geq 2\sigma$ after Euclid DR1 systematics corrections:

ID	Condition
KC1	$ E_G(\ell \approx 66) - E_G^{\Lambda\text{CDM}} < 1\%$ at 2σ (including systematics) — i.e., consistent with zero shift versus the predicted -3.98% ; falsifies the EFC benchmark coupling.
KC2	Any of the five sealed observables (P1–P5) shows an <i>opposite-sign</i> shift relative to ΛCDM at $\geq 2\sigma$ after window/convolution corrections.
KC3	Joint Euclid DR1 growth and lensing sectors shift with the <i>same sign</i> near $k \approx 0.05 h/\text{Mpc}$ (or $\ell \approx 66$) at $\geq 2\sigma$, violating EFC coherence.
KC4	Independent reproduction with hi_class (using the supplied <code>efc_logistic</code> implementation) reveals Horndeski stability violation ($c_s^2 \leq 0$ or $Q_S \leq 0$) on $0 < z < 3$ at the stated benchmark.

5 No Post-Hoc Adjustment

All predictions in this document are fixed prior to the release of Rubin DP2, Euclid DR1, and $\text{SO} \times \text{Euclid}$ cross-correlations. No parameter tuning, refitting, or reinterpretation will be performed after data release.

6 Regime Consistency Note (RCMP)

The EFC predictions arise from regime-consistent dynamics (DOI 31222900), where observables reflect transitions between measurement regimes rather than parameter adjustments within a single regime. This principle explains $\mu < 1$ and $\eta > 1$ as cross-regime effects. RCMP is included as interpretive context and does not introduce additional free parameters.

7 Scope

This pre-registration applies specifically to:

- Weak lensing and clustering: Euclid DR1 (October 21, 2026).
- Growth measurements: Rubin DP2 (July–September 2026).
- E_G observable: $\text{SO} \times \text{Euclid}$ cross-correlation (~ 2027).

8 Final Statement

This document constitutes a complete, frozen set of predictions for the EFC model prior to the availability of the targeted observational datasets. All numerical values and constraints are derived from independently archived and timestamped sources (DOIs listed above), sealed by SHA-256 hash, and are not subject to modification.

References (DOIs):

- 32010399 — Pre-Registered Predictions for the EFC Model (this document).
 - 31876324 — EFC Relativistic Action: Field Equations, Perturbation Theory, and Extraction of μ , Σ , η .
 - 31990053 — EFC Euclid DR1 Boltzmann Pre-Registration (sealed).
 - 31332730 — $f\sigma_8$ Leave-One-Out Signal Analysis.
 - 31230703 — Regime-Transition Fit to DESI DR2 BAO with Cross-Validation.
 - 31222900 — The Regime-Consistent Measurement Principle (RCMP).
- All DOIs resolve via <https://doi.org/10.6084/m9.figshare.NNNNNNN>.

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